

Laboratory 2 (100 Points)

Assignment 1 (20 Points)

Implement a logic circuit for the two-bit comparator. This comparator has two two-bit inputs A ($A = A_1A_0$) and B ($B = B_1B_0$) and produces one output. The output should be 1 if A is greater than B, otherwise the output should be 0.

- Construct the truth table for the two-bit comparator
- Write the sum of product form of the function from the truth table.
- Simplify the function as much as possible
- Design a logic diagram for the simplified function
- Run the simulation of the logic diagram using Simulation Waveform Editor tool.

a. Construct the truth table for the two-bit comparator

A1	A0	B1	B0	A_greater_than_B
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

A > B (A = 1, B = 0)

A > B (A = 2, B = 0)

A > B (A = 2, B = 1)

A > B (A = 3, B = 0)

A > B (A = 3, B = 1)

A > B (A = 3, B = 2)

b. Write the sum of product form of the function from the truth table.

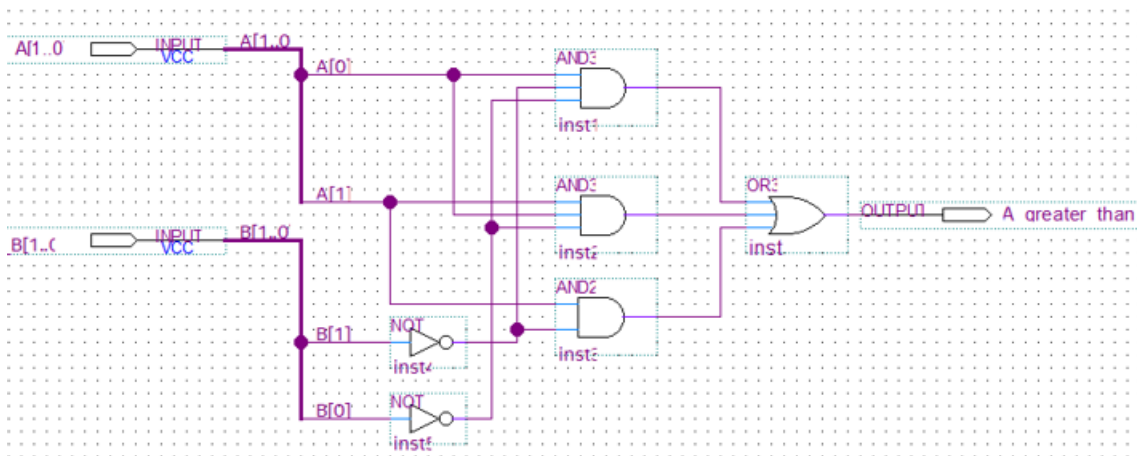
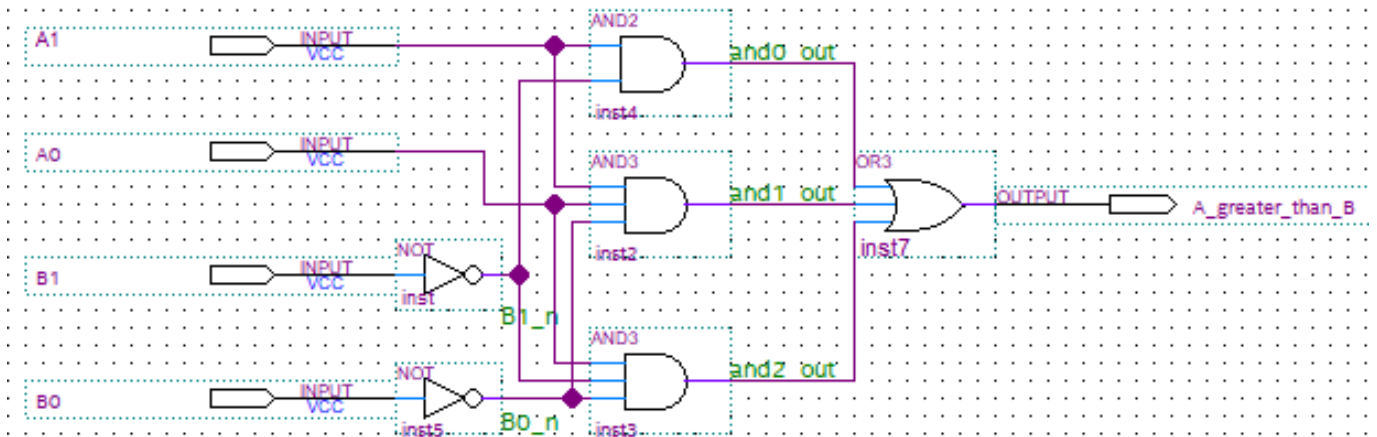
$$A_greater_than_B = A_1'A_0B_1'B_0' + A_1A_0'B_1'B_0' + A_1A_0'B_1'B_0 + A_1A_0B_1'B_0' + A_1A_0B_1'B_0 + A_1A_0B_1B_0'$$

c. Simplify the function as much as possible

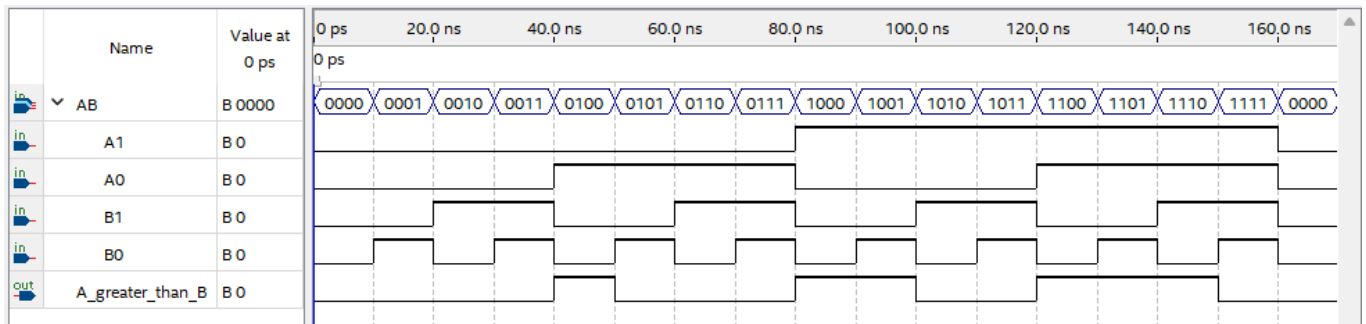
		B1B0			
		00	01	11	10
A1A0	00				
	01	1			
	11	1	1		1
	10	1	1		

$$A_greater_than_B = A0B1'B0' + A1A0B0' + A1B1'$$

d. Design a logic diagram for the simplified function



e. Run the simulation of the logic diagram using Simulation Waveform Editor tool



Assignment 2 (20 Points)

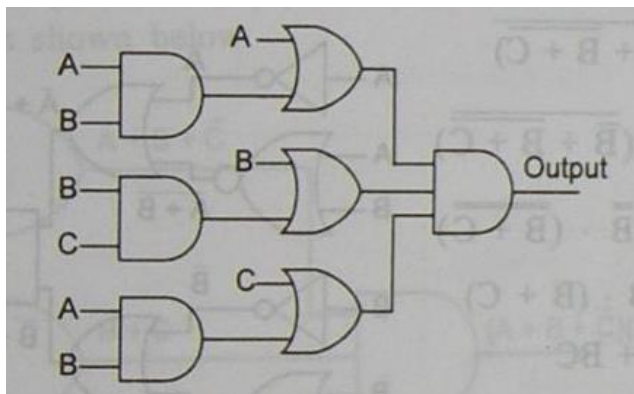
Design a logic diagram for the following Boolean expressions and verify the waveform with its simplified Boolean expression (as much as possible) truth table.

- a) $AC + AC' + A'C$
- b) $AB + A(B+C)'$
- c) $AB'C + B + BD' + ABD' + A'C$

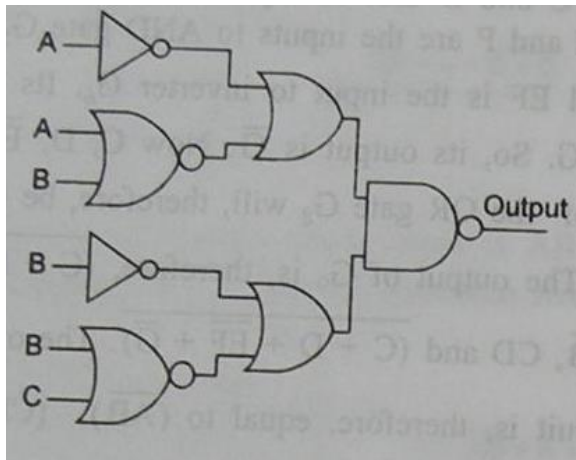
Assignment 3 (20 Points)

Write the Boolean expressions for the following a logic diagram. Then verify the waveform with its simplified Boolean expression (as much as possible) truth table.

(a)



(b)



Assignment 4 (20 Points)

Reduce the following expression in SOP and POS forms using 3 variable K-map and design the logic diagram then verify them with their truth tables and waveforms.

a) $\Sigma m(1, 3, 4, 5, 6, 7)$

b) $\Pi M(3,5,7)$

Assignment 5 (20 Points)

Reduce the following expression in SOP and POS forms using 4 variable K-map and design the logic diagram then verify them with their truth tables and waveforms.

a) $\Sigma m(0,1,2,3,4,6,9,11)$

b) $\Pi M(1,3,4,5,6,9,10,11,12,13,14)$